Distilling the Prompt Pathway

NeuroVestigo — closing the frame-0 conditioning gap in SAM 2 video propagation

What

SAM 2 segments the first frame of a clip with the ground-truth mask supplied as a *dense prompt*, and every later frame with no prompt at all — memory cross-attention is their only spatial context. The two regimes are not equally good. We train the no-prompt pathway to reproduce what the prompted pathway already achieves, using the model itself as the teacher.

Why — the measurement that motivates it

Per-frame validation metrics, shipped 16-frame model, best epoch (110-epoch run, 5 validation recordings):

	frame 0 (prompted)	frames 1–15 (memory only)	gap
IoU	0.9466	∼0.830	−12%
Betti-0 error	0.080	∼0.70	∼ **9**×

Two facts do the work. First, the topology gap is an order of magnitude larger than the region gap: losing the prompt does not blur the mask, it **fragments neurites**. Second, frames 1→15 drift only **0.017 in total** — essentially flat. So this is a *single conditioning step*, not accumulating propagation error. There are only two regimes, prompted and not, and every unprompted frame sits at the same level. Improving that pathway therefore lifts all fifteen frames together; no new cliff appears at frame 2.

The decoder can produce a 0.947 mask of these images. It simply cannot reach that without a spatial prior.

How

(a) Teacher with privileged information. At $k = 4$ randomly sampled frames per clip, decode a second time with that frame's ground truth as the dense prompt. Same weights, same image features (the encoder runs once); the teacher differs only in receiving information unavailable at inference. Its output is detached.

(b) Prompt dropout. Frame 0's prompt is withheld with probability p, so the conditioning frame is sometimes segmented from features alone — the regime every other frame is always in. Ramped over E_w epochs so an early bad frame-0 prediction cannot poison a clip's memory bank.

Math

Let f_θ be the network, x_t frame t, m_t its ground-truth mask, $M_{<t}$ the memory bank, and π the dense prompt. Teacher and student differ *only* in π:

$$\underbrace{\tilde{y}_t = f_\theta(x_t,\ \pi = m_t)}_{\text{teacher: privileged}} \qquad \underbrace{\hat{y}_t = f_\theta(x_t,\ \pi = \varnothing,\ M_{<t})}_{\text{student: inference-time}}$$

$$\mathcal{L}_{\text{KD}} = \lambda_1\, \mathcal{L}_{\text{Dice}}(\hat{y}_t,\ \text{sg}[\sigma(\tilde{y}_t)]) + \lambda_2\, \mathcal{L}_{\text{cbDice}}(\hat{y}_t,\ \text{sg}[\sigma(\tilde{y}_t)])\,, \qquad \lambda_1 = 1.0,\ \lambda_2 = 0.5$$

with sg[·] the stop-gradient, so the loss only ever moves the student. The target is $\sigma(\tilde{y}_t) \in [0, 1]$, *soft*: it encodes which pixels the model finds genuinely ambiguous, which a binary mask cannot. Dice accepts this directly, $\mathcal{L}_{\text{Dice}}(a, b) = 1 - \frac{2\sum \sigma(a) b + \epsilon}{\sum \sigma(a) + \sum b + \epsilon}$. Prompt dropout at epoch e:

$$\pi_0 = \begin{cases} m_0 & \text{w.p. } 1 - p_e \\ \varnothing & \text{w.p. } p_e \end{cases} \qquad p_e = p \cdot \min(1,\ e/E_w)\,, \qquad p = 0.4,\ E_w = 40.$$

Full objective (weights as shipped):

$$\mathcal{L} = 3.0\, \mathcal{L}_{\text{focal}} + 2.0\, \mathcal{L}_{\text{Dice}} + 0.3\, \mathcal{L}_{\text{cbDice}} + 0.3\, \mathcal{L}_{\text{sv}} + 1.0\, \mathcal{L}^{\text{KD}}_{\text{Dice}} + 0.5\, \mathcal{L}^{\text{KD}}_{\text{cbDice}}$$

Provenance — what is borrowed, what is ours

Component	Source
Training a student on a teacher's *soft* outputs	Hinton, Vinyals & Dean, arXiv:1503.02531
Teacher holding information the student will never have	Lopez-Paz, Bottou, Schölkopf & Vapnik, ICLR 2016, arXiv:1511.03643
Withholding the prompt during training (exposure bias)	Bengio, Vinyals, Jaitly & Shazeer, NeurIPS 2015, arXiv:1506.03099
Teacher = the same weights, differing only in conditioning	**ours** — not a second network, not a deeper layer, not a prior generation
Using **cbDice** as the distillation target	**ours** — follows from the 9× topology gap above; no paper suggests it
The frame-0 conditioning gap as a measured defect	**ours** — the per-frame IoU/Betti split that motivates all of it
The loss functions themselves	the repo's existing Dice / cbDice. *Not* from the KD literature: Hinton's objective is temperature-scaled KL on softmax, which does not transfer to binary segmentation

Cost and how it will be judged

No new parameters: 5,557,250 trainable of 39,876,930 (13.9%), unchanged. The image encoder is shared, so each teacher frame costs one prompt-encoder + mask-decoder pass; with $k = 4$ of 16 frames the measured overhead is ≈30–40% per step.

Read Betti-0 before IoU. The mechanism predicts topology moves first, since that is where the 9× gap is. A run that pulls validation b_0 from ∼0.70 toward ∼0.30 while IoU barely moves is the intended result, not a null one. *Status: training in progress; no result yet.*